

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

October 7, 2026

Recommendation Systems: Spotify

Pasick (2022) on Spotify's "Discover Weekly":

- A hybrid system:
 - Collaborative filtering (using four billion+ user playlists)
 - Natural language processing (to "read" music blogs, etc.)
 - Deep learning (incorporates audio features)
- Hybrid model allows recommendations of new songs ("cold starts")
- System finds the gap between your taste profile and what similar listeners have discovered



Recommendation Systems: Netflix

Chandrashekar et al. (2017) on Netflix:

- Netflix: Not just *what* to present, but *how* (different users respond to different artwork for the same title)
- “Contextual bandits”: continuously updates artwork based on user behavior
- Challenges: cold-starts, attribution, and scale



Ethical Questions

1. What Are They Maximizing?
2. Is Personalization Power?
3. How Informed Is Consent?
4. Whither Filter Bubbles?
5. What About The Creators?

Design A Recommender